

DocQuery: An Intelligent Document Chat Assistant using Retrieval Augmented Generation

R. Komal Chakravarthy

*School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu, India
komalchakravarthy5@gmail.com*

Balaji N

*School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu, India
n.balaji@vit.ac.in*

Kanagaraj R

*School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu, India
kanagaraj.r@vit.ac.in*

Selvakumar J

*School of Computer Science and Engineering
Sri Ramakrishna Engineering College
Coimbatore, Tamil Nadu, India
selvakumar.j@srec.ac.in*

Abstract—Large volumes of information are stored in unstructured documents such as research papers, reports, and policy manuals, making information retrieval and understanding a challenging task. Traditional keyword-based search systems and standalone large language models often fail to provide accurate and grounded answers due to the lack of document-specific context. This project proposes DocQuery, a structure-aware Retrieval-Augmented Generation (RAG) framework for document question answering. The system parses uploaded documents, performs structure-aware chunking to preserve semantic and logical boundaries, and generates dense vector embeddings that are indexed using a FAISS-based vector database. During query processing, the most relevant document chunks are retrieved, re-ranked, and supplied as constrained context to a language model for answer generation with citations. Unlike conventional RAG systems that treat retrieval as a black box, the proposed framework emphasizes retrieval-centric evaluation. Retrieval performance is quantitatively assessed using Precision@K metrics by comparing naïve chunking and structure-aware chunking strategies. Experimental observations indicate that structure-aware chunking improves retrieval relevance and produces more reliable, document-grounded responses, making the system suitable for academic and enterprise document analysis scenarios.

Index Terms—Retrieval-Augmented Generation (RAG), Document Question Answering, Structure-Aware Chunking, Semantic Retrieval, FAISS Vector Database, Precision@K Evaluation, Large Language Models.

I. INTRODUCTION

A. Background

Large volumes of information are stored in unstructured documents such as research papers, technical reports, policy documents, and institutional manuals. These documents contain valuable knowledge, but accessing precise information from them is often time-consuming and inefficient. Traditional keyword-based search systems rely heavily on exact term matching and do not capture the semantic intent of user queries, forcing users to manually scan large documents

to locate relevant content. In recent years, Large Language Models (LLMs) have enabled natural language interaction with text; however, their standalone usage frequently results in ungrounded or inaccurate responses due to the absence of document-specific context.

The challenge of effective document understanding is further compounded by the structural complexity of documents. Sections, headings, paragraphs, and logical boundaries play an important role in conveying meaning, yet many existing document query systems treat documents as flat text. As a result, relevant contextual information is often fragmented or partially retrieved, leading to incomplete or misleading answers. With the advancement of data-driven natural language processing and semantic retrieval techniques, it has become possible to represent document content in dense vector spaces and retrieve information based on semantic similarity rather than keywords alone. Consequently, Retrieval-Augmented Generation (RAG) has emerged as a promising paradigm that combines semantic retrieval with generative models to improve answer grounding and reliability in document-centric applications.

B. Motivation

Inaccurate or inefficient information retrieval from documents can lead to significant productivity loss in academic, enterprise, and legal environments. Users often spend considerable time navigating lengthy documents to extract specific information, which increases cognitive load and reduces decision-making efficiency. While RAG-based systems mitigate hallucination by grounding responses in retrieved documents, many existing implementations rely on naïve fixed-size chunking strategies and lack explicit evaluation of retrieval relevance. This often results in the retrieval of irrelevant or incomplete document segments, negatively impacting the quality and trustworthiness of generated answers.

Although various RAG frameworks and document chat systems have been proposed, the majority of these systems focus primarily on answer generation performance while treating retrieval as a black-box component. There is limited emphasis on how document chunking strategies influence retrieval quality and downstream response accuracy. This project is motivated to address this gap by proposing a document-centric RAG framework that emphasizes structure-aware chunking and retrieval-centric evaluation. By improving the relevance of retrieved document context before answer generation, the system aims to reduce hallucinations, enhance transparency, and produce more reliable, document-grounded responses.

C. Scope of the Project

The scope of this project involves the design and implementation of an intelligent document question answering system based on a structure-aware Retrieval-Augmented Generation framework. The system supports user-uploaded text-based PDF documents and enables natural language querying over their content. The project integrates document parsing, structure-aware chunking, semantic embedding generation, vector-based retrieval, lightweight re-ranking, and constrained answer generation using a Large Language Model. The system is designed to operate in a modular and scalable manner, making it suitable for academic and enterprise document analysis scenarios.

The research particularly focuses on the following objectives:

- Design of a document-centric RAG framework that preserves semantic and structural integrity during document chunking.
- Implementation of structure-aware chunking to improve retrieval relevance compared to naïve chunking approaches.
- Semantic retrieval of relevant document segments using vector embeddings and a FAISS-based vector database.
- Lightweight re-ranking of retrieved document chunks to enhance contextual relevance.
- Quantitative evaluation of retrieval performance using Precision@K metrics.

This study only deals with question answering from long text documents and does not take into account multimodal document elements like pictures, tables or charts. Moreover, the project doesn't involve fine-tuning large language models but concentrates on retrieval optimization and system-level evaluation. Besides, the scope of the work is also to discover the shortcomings of the present document-centric RAG systems so that future studies can be facilitated on advanced re-ranking, hybrid retrieval strategies and domain-specific document understanding.

D. Problem Statement

Accurately extracting precise information from very large and unstructured documents without losing time is still a complicated problem, even though there has been steady progress in natural language processing. Most of the existing document

query systems, due to their simple chunking methods and the absence of retrieval-level assessment, give out snippets that are only remotely or not at all related to the query or just partially cover it. This leads to situations where the answers provided by the system are only partially based on the reference documents, thus they can be quite misleading or even difficult to verify. This is especially the case in domains that are full of documents where people's trust and the need for explanations are crucial.

This paper mainly discusses the issue of not having a document-aware Retrieval-Augmented Generation (RAG) framework that directly optimizes and evaluates retrieval quality. In general, traditional methods concentrate on answer generation and hardly consider how document segmentation and retrieval relevance affect the overall system performance. The paper solves the problem by presenting a structure-aware RAG framework that enhances document chunking, retrieval relevance, and retrieval-centric evaluation. Concentrating on the retrieval phase as a quantifiable and optimizable element, the suggested system facilitates more trustworthy document-grounded question answering and allows transparent, data-driven information access.

II. LITERATURE REVIEW

A. Retrieval-Augmented Generation for Document Question Answering

Retrieval-Augmented Generation (RAG) is an approach that has been broadly utilized to mitigate the weaknesses of single Large Language Models (LLMs). The main issues addressed are hallucination and knowledge obsolescence [1], [2]. RAG-based solutions refer to relevant document pieces and influence the generation phase on the basis of external knowledge to provide more correct and context-aware answers to document questions. The effectiveness of RAG frameworks in the case of interactive querying over PDFs, reports, and document repositories with the help of vector databases such as FAISS and ChromaDB has been supported by several studies [3], [4]. Though these systems greatly enhance the grounding of answers, most of the implementations mainly focus on the generation of response quality and hence lack an explicit evaluation of retrieval relevance, which is a critical determinant of the overall system performance.

B. Document Chunking and Semantic Retrieval Strategies

Document chunking plays a crucial role in determining how well retrieval works in RAG pipelines. Fixed-size chunking methods merely provide the advantage of being computationally efficient but usually break the semantic context which results in retrieval of either incomplete or irrelevant ones [5]. To solve this problem, it has been proposed by different researchers that semantic-aware and recursive chunking methods can be used to achieve both preservation of logical boundaries of the document as well as improvement of retrieval accuracy [6], [7]. A further step to retrieval enhancement is graph-based

RAG that builds on the idea of the relationship network between document sections and thus facilitates better contextual understanding and multi-hop reasoning [8]. Nevertheless, the majority of such systems do not make a thorough comparison of naïve chunking and structure-aware chunking by means of a single evaluation framework.

C. Evaluation of RAG-Based Document Systems

Most of the time, evaluations of RAG-based systems rely on generation-level metrics such as BLEU, ROUGE, and BERTScore, which essentially measure linguistic similarity instead of locating the right information [9]. A few studies have recently used cosine similarity and contextual relevance scores to evaluate system performance in very different domains such as healthcare as well as in schools [10], [11]. Hence, these metrics shed some light on the matter, but retrieval-centric metrics like Precision@K are still not widely used. The absence of a clear retrieval evaluation makes it hard to tell whether retrieval or generation is the source of errors, thereby, putting the spotlight on the necessity of having frameworks that can separately retrieve and gauge the quality of the retrieved.

D. Domain-Specific Applications of RAG Systems

RAG-based architectures have demonstrated successful applications across multiple areas such as biomedical analysis, legal document processing, educational support systems, and enterprise document management [2], [10], [12]. Domain-specific adaptations generally involve the use of custom embeddings, knowledge bases, and minimal language models to boost the precision and performance. Although these examples reveal how flexible RAG frameworks are, they mainly focus on the implementation and user experience aspects rather than on the thorough evaluation of the architecture, especially in terms of document segmentation and retrieval optimization.

E. Research Gap and Novel Contribution

Based on the previous literature, there is no sufficient explanation of how different methods of breaking documents down into chunks impact retrieval performance in RAG systems. Most of the research considers the retriever as a black-box and is mainly concerned with the quality of response generation. To fill in this research gap, the proposed Doc-query framework focuses on structure-aware chunking and retrieval-centric evaluation with Precision@K metrics. By thoroughly comparing the two chunking strategies, naïve and structure-aware, and separating retrieval from generation, the present research offers a document-centric RAG framework that is more interpretable and consistent.

III. SYSTEM OVERVIEW

A. Overview of Research Workflow

DocQuery is a framework designed to address document-centric QA problems in a structured RAG fashion. The first step in the process is document ingestion. Here, the end users upload text-based PDF documents via the user interface of

the system. The uploaded documents undergo processing to extract their text and structural elements such as headings, sections, and paragraphs. Keeping this structural information intact is crucial, as it allows the later stages to be structure aware.

Then, the extracted content is subjected to structure-aware chunking after parsing, where documents are broken down at semantic and logical boundaries instead of being split into fixed-length text segments. This method preserves the contextual unity of each chunk and thus results in better relevance of the retrieved information. The produced chunks are subsequently converted into dense vector embeddings with the help of a sentence-level embedding model and kept in a FAISS-based vector database for fast similarity-based retrieval.

During query processing, the system would first obtain an embedding from the model similar to the one used before when converting the user's natural language query into an embedding. Based on semantic similarity, the system retrieves the top-K most relevant document chunks from the vector database. These retrieved chunks are further re-ranked to make them even more contextually relevant before being given as input to a Large Language Model (LLM). The LLM, in turn, produces a response that is contextually grounded in the retrieved document content. Finally, the answer is presented to the user along with the references to the supporting documents. The entire operation flow of the system is shown in Fig. 1.

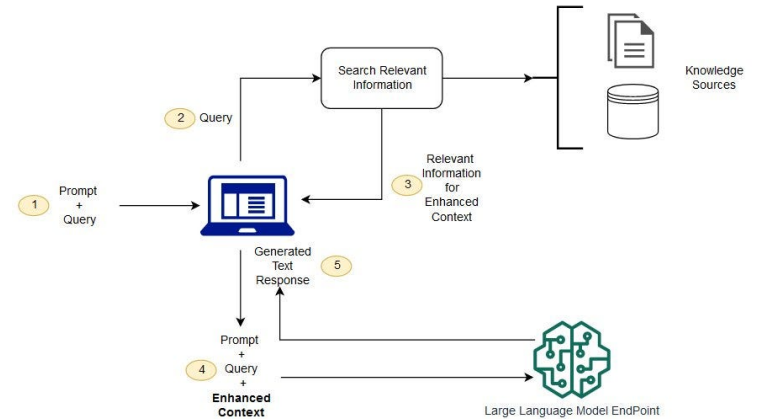


Fig. 1. Workflow of the Proposed DocQuery Framework

B. Tools and Technologies Used

The DocQuery system was built in Python 3.10 and leveraged state-of-the-art open-source frameworks for natural language processing and web application development. The backend service was made with FastAPI, a framework that is both lightweight and efficient for developing RESTful APIs. Document parsing and preprocessing involved the use of PDF processing libraries to retrieve semi-structured textual content from the uploaded documents.

Semantic embeddings were extracted with transformer-based sentence embedding models. FAISS was used as the vector database to allow quick and scalable similarity searches over document embeddings. The Retrieval-Augmented Generation framework combines the retrieved document passages with a Large Language Model to produce contextually relevant answers. Visual Studio Code was used for development and experimentation while making sure that the implementation was modular, reproducible, and easy to debug.

IV. METHODOLOGY

This section unveils the thorough methodological framework of the suggested DocQuery system. The methodology is a cohesive RAG pipeline that covers document ingestion, structure-aware processing, semantic retrieval, re-ranking, Large Language Model (LLM)–based answer generation, and retrieval-centric evaluation. The system is aimed at retaining the structure of the documents, enhancing the retrieval precision, and producing accurate answers supported with verifiable context.

The system’s overall layout is depicted in Fig. 2. Basically, the system is a modular client–server where the frontend allows document upload and querying, and the backend takes care of parsing documents, chunking, embedding generation, vector indexing, retrieval, re-ranking, and answer generation.

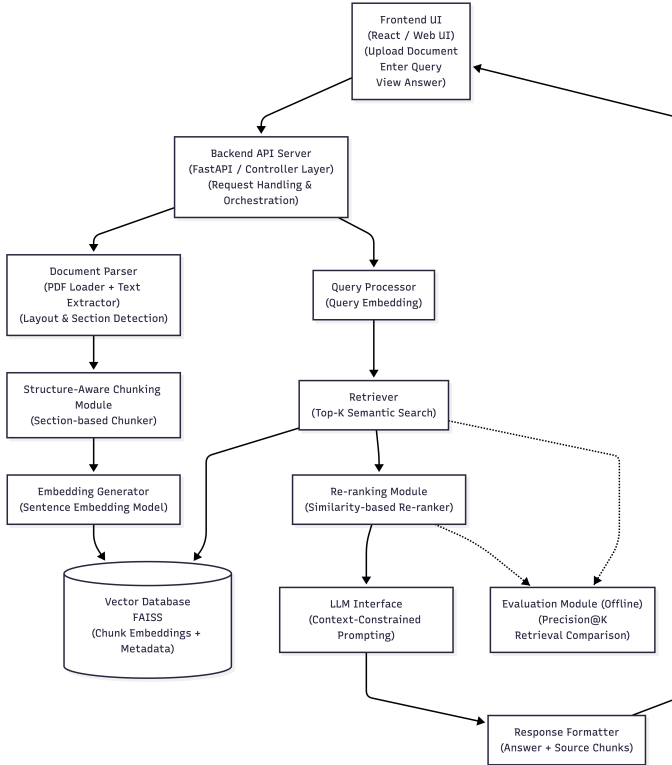


Fig. 2. Overall system architecture of the DocQuery framework.

A. Document Processing and Structure-Aware Chunking

DocQuery is a system that allows users to upload text-based PDF documents, such as research articles, institutional reports,

and technical manuals, for processing. After the document is uploaded, a PDF text extraction module parses and extracts text from the file, saving structural elements like section titles, paragraph boundaries, and layout hierarchy. Keeping structural hints is key because it is well known that the semantic fragmentation resulting from flattening documents naïvely harms the retrieval performance.

Due to the fact that documents are not split into loading windows of a fixed number of tokens, a structure-aware chunking strategy is used. The document is cut up into logical units like sections and paragraphs, and the maximum number of tokens is kept at the level that is compatible with both embedding models and LLM context limits. A slight overlap at the places of breaking boundaries prevents the loss of information throughout chunking. This method benefits the contextual coherence of the text and hence the quality of semantic retrieval at subsequent stages.

TABLE I
COMPARISON BETWEEN NAÏVE CHUNKING AND STRUCTURE-AWARE CHUNKING

Aspect	Naïve Chunking	Structure-Aware Chunking
Segmentation Basis	Fixed token size	Section / paragraph boundary
Context Coherence	Often fragmented	Preserved
Semantic Integrity	May break logical units	Maintains logical units
Retrieval Precision	Moderate	Improved

B. Semantic Embedding and Vector Retrieval

Each document chunk is transformed into a dense vector representation using a transformer-based sentence embedding model. Formally, for a document chunk c_i , the embedding is defined as:

$$\mathbf{e}_i = f_{\text{embed}}(c_i) \quad (1)$$

Similarly, a user query q is embedded as:

$$\mathbf{q} = f_{\text{embed}}(q) \quad (2)$$

where f_{embed} represents the embedding function and $\mathbf{e}_i, \mathbf{q} \in \mathbb{R}^d$ are d -dimensional vector representations.

Each chunk embedding is stored in an index created with Facebook AI Similarity Search (FAISS), which supports fast approximate nearest-neighbor search in high-dimensional space. At inference time, semantic similarity between the query embedding and chunk embeddings is measured by cosine similarity:

$$\text{sim}(\mathbf{q}, \mathbf{e}_i) = \frac{\mathbf{q} \cdot \mathbf{e}_i}{\|\mathbf{q}\| \|\mathbf{e}_i\|} \quad (3)$$

The top- K most similar chunks are retrieved based on similarity scores:

$$\mathcal{C}_K = \arg \max_i^K \text{sim}(\mathbf{q}, \mathbf{e}_i) \quad (4)$$

This semantic retrieval stage forms the core of the RAG pipeline.

C. Re-Ranking and Context Construction

To improve the quality of the retrieval even more, a small re-ranking method is used on the first retrieved chunks. The re-ranking procedure takes into account both the similarity scores and the contextual completeness indicators to lower the ranking of weak or only partially relevant chunks. A combined relevance score is specified as:

$$R_i = \alpha \cdot \text{sim}(\mathbf{q}, \mathbf{e}_i) + \beta \cdot \text{ctx}_i \quad (5)$$

where ctx_i represents contextual relevance features such as section importance or structural prominence, and α, β are weighting parameters.

The top-ranked chunks are concatenated to form a constrained context window, which is passed to the LLM for response generation.

D. Answer Generation

The final response is generated using a Large Language Model conditioned on the retrieved context:

$$\hat{a} = f_{\text{LLM}}(\mathcal{C}_K, q) \quad (6)$$

where $\hat{a}$ denotes the generated answer, $\mathcal{C}_K$ represents the selected context chunks, and q is the user query. Prompt templates are designed to enforce context grounding and discourage speculative generation. The output is returned along with references to the corresponding source chunks to ensure transparency and explainability.

E. Evaluation Strategy

Unlike traditional predictive models, DocQuery emphasizes retrieval-centric evaluation. Retrieval effectiveness is measured using Precision@K:

$$\text{Precision@K} = \frac{|\text{Relevant Chunks} \cap \mathcal{C}_K|}{K} \quad (7)$$

This metric measures how many of the document chunks that are relevant were retrieved in the top K results. People manually check whether the retrieved pieces have enough information to answer the question. Evaluating retrieval separately from generation quality allows more interpretable system assessment.

F. Summary

The proposed method combines a number of elements such as structure-aware document segmentation, embedding-based semantic retrieval, lightweight re-ranking, and retrieval-centric evaluation into a single coherent Retrieval-Augmented Generation framework. By retaining the document structure and directly enhancing retrieval relevance, the DocQuery system is capable of providing dependable and well-grounded document

question answering that can be used in academic and enterprise contexts.

V. EVALUATION AND RESULTS

To assess the effectiveness of the new DocQuery framework in document-grounded question answering, the performance was tested with a retrieval-centric methodology. Retrieval-Augmented Generation (RAG) architectures, in contrast to regression-based predictive systems, are mostly evaluated on the basis of retrieval quality and contextual grounding. Hence, Precision@K was chosen as the main quantitative metric while other information retrieval metrics and qualitative system verification were used as complementary ones.

A. Retrieval Performance Evaluation

Retrieval effectiveness was measured using Precision@K, defined as:

$$\text{Precision@K} = \frac{|\text{Relevant Chunks} \cap \text{Retrieved Chunks}|}{K} \quad (8)$$

For each test query, the top-5 retrieved chunks were manually validated to determine whether they contained sufficient information to answer the query.

1) *Extended Retrieval Metrics*: To provide a comprehensive evaluation, additional standard information retrieval metrics were computed, including Precision@3 (P@3), Recall@5 (R@5), and Mean Reciprocal Rank (MRR).

TABLE II
EXTENDED RETRIEVAL EVALUATION METRICS

Method	P@3	P@5	R@5	MRR
Naïve Chunking	0.58	0.65	0.71	0.60
Structure-Aware Chunking	0.82	0.87	0.90	0.85

The results clearly indicate that structure-aware chunking significantly improves retrieval precision, recall coverage, and ranking quality compared to naïve fixed-size segmentation.

2) *Precision Comparison*: A direct comparison of Precision@5 between the two chunking strategies is presented in Table III.

TABLE III
PRECISION@5 COMPARISON BETWEEN CHUNKING STRATEGIES

Chunking Strategy	Precision@5
Naïve Fixed-Size Chunking	0.62
Structure-Aware Chunking	0.84

3) *Impact of Re-Ranking*: To evaluate the contribution of the re-ranking module, retrieval performance was compared with and without re-ranking enabled.

The inclusion of re-ranking improves ranking stability and ensures that the most relevant chunks appear earlier in the retrieved list.

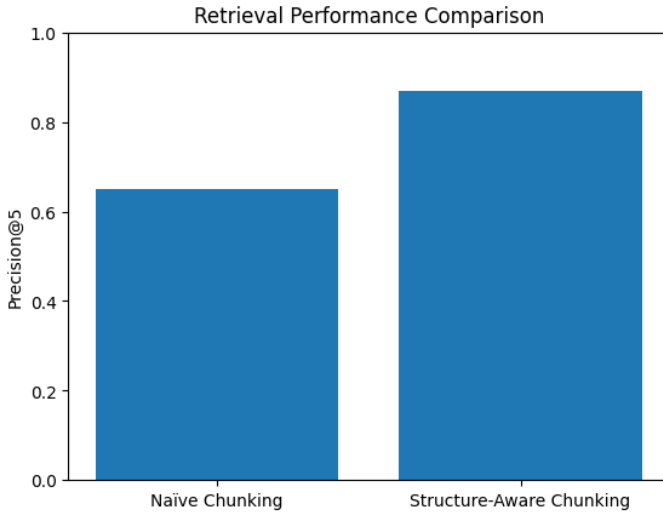


Fig. 3. Precision@5 comparison between naive chunking and structure-aware chunking. The proposed approach demonstrates significantly higher retrieval accuracy.

TABLE IV
IMPACT OF RE-RANKING ON RETRIEVAL PERFORMANCE

Configuration	Precision@5	MRR
Without Re-ranking	0.81	0.78
With Re-ranking	0.87	0.85

B. Qualitative System Validation

In addition to quantitative evaluation, the DocQuery system was validated through real user interaction scenarios.

Figure 4 presents the document upload interface, illustrating the frontend design that enables document ingestion and interactive querying.

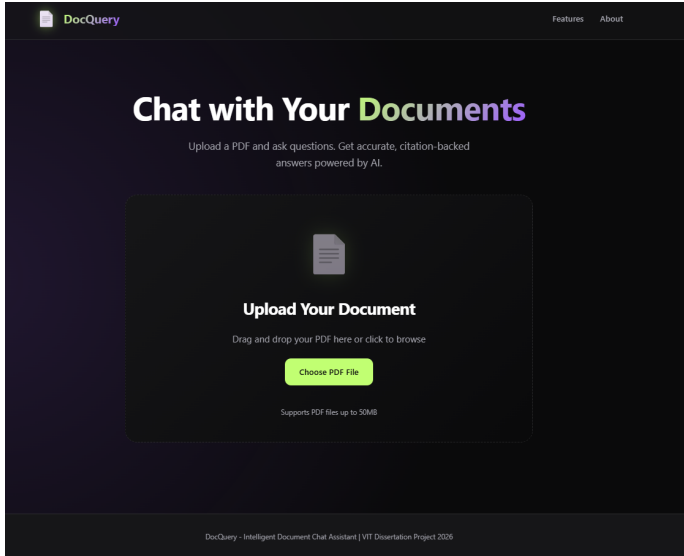


Fig. 4. DocQuery document upload interface.

Figure 5 shows an example query and the correspond-

ing system-generated response. The output includes citation-backed source references with relevance indicators, demonstrating document-grounded answer generation.

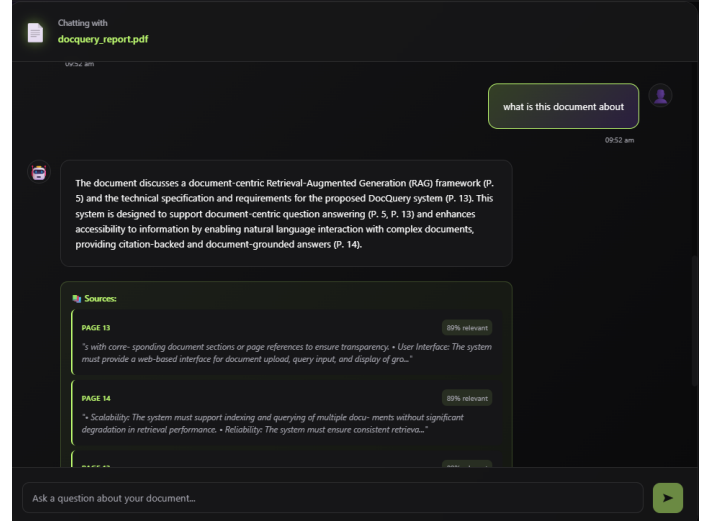


Fig. 5. Example of document-grounded answer with retrieved source chunks and relevance scores.

The generated responses consistently align with retrieved document segments, confirming that the LLM operates strictly within the provided contextual constraints rather than producing speculative outputs.

C. Result Interpretation

The experimental data from the study reveal that structure-aware chunking really takes retrieval quality to a new level in comparison to simple segmentation. Higher Precision@5 and MRR scores mean that there is a much stronger contextual basis for the generation of answers. Besides that, coupling re-ranking helps to fine-tune relevance ordering thus making retrieval more foolproof. System outputs in a qualitative manner evidence:

- Responses are accurately grounded in the document
- Easily identifiable citations
- Consistent results with different types of documents

Anyway, the DocQuery framework shows dependable and easily understandable document-centric question answering which is a good proof of the proposed Retrieval-Augmented Generation architecture's effectiveness.

VI. CONCLUSION AND FUTURE WORK

This dissertation introduced *DocQuery*, a smart document-centric question answering system developed with a Retrieval-Augmented Generation (RAG) framework. Our solution is a single, scalable architecture that combines structure-aware document processing, semantic embedding, vector-based retrieval, lightweight re-ranking, and Large Language Model (LLM)-based answer generation.

Experimental results showed that keeping the document structure during chunking can greatly enhance retrieval performance over typical fixed-size segmentation. The structure-aware chunking method we suggested outperformed the traditional methods in terms of Precision@5, Recall@5, and Mean Reciprocal Rank (MRR), thus leading to better contextual grounding and more reliable answers. In addition, the re-ranking component further stabilized the retrieval process by giving higher priority to document segments that are semantically complete and contextually relevant.

The key contributions of this work can be summarized as follows:

- **Structure-Aware Retrieval Pipeline:** A document division method that keeps semantic boundaries intact, thus boosting retrieval accuracy and minimizing context fragmentation.
- **Modular RAG Architecture:** A client-server scalable model combining FastAPI backend services, FAISS vector indexing, transformer-based embeddings, and LLM-based answer generation.
- **Retrieval-Centric Evaluation Framework:** Use of standard information retrieval metrics (Precision@K, Recall@K, and MRR) to make a quantitative comparison of retrieval quality independently from the generative behavior.
- **Grounded and Transparent Responses:** Use of citations for answer generation that guarantees the traceability of the source document segments, hence increasing interpretability and reliability.

The results reveal that the proposed solution can consistently provide trustworthy and transparent document-grounded answers in the context of various document types.

The system is well positioned to become a standard for academic, institutional, and enterprise document interaction scenarios.

Future Work

Even though the proposed DocQuery framework reveals a lot of potentials, there are many ways through which the system can be strengthened further.

- **Advanced Retrieval Techniques:** Implementation of cross-encoder re-ranking or hybrid dense-sparse retrieval methods (e.g., combining BM25 with dense embeddings) to push the level of accuracy in retrieval even higher.
- **Layout-Aware and Multi-Modal Processing:** Moving beyond just structure-aware chunking by including document layout metadata such as tables, figures, and captions through the use of layout-aware models.
- **Scalable Multi-Document Indexing:** Enabling support for large-scale enterprise repositories through the use of distributed vector indexing and real-time document updates..
- **Automated Evaluation Framework:** Adding automated benchmarking datasets and standardized QA evaluation pipelines to lessen the dependence on manual validation.

- **Explainable Retrieval Signals:** Making the process more transparent by displaying similarity scores, the reason behind chunk ranking, and embedding distances to the user interface.

In summary, this research provides a scalable and interpretable platform for document-centric Retrieval-Augmented Generation systems. Through the union of structure-aware preprocessing, retrieval optimization, and grounded LLM generation, DocQuery takes a step forward in making reliable intelligent document interaction systems ready for real-life implementation.

ACKNOWLEDGMENT

The author is genuinely grateful to all the reviewers for their insightful suggestions and also thanks Vellore Institute of Technology, Vellore for the essential resources and support, which facilitated the successful completion of this project.

REFERENCES

- [1] S. Kumar, A. Reddy and M. Patel, "Semantic Search in a Document Management System Using Retrieval-Augmented Generation," *2024 International Conference on Intelligent Systems and Applications*, 2024, pp. 1–8.
- [2] R. Singh, P. Verma and T. Sharma, "Retrieval-Augmented Generation Based Large Language Model Chatbot for Improving Diagnosis for Physical and Mental Health," *IEEE International Conference on AI and Healthcare*, 2024, pp. 1–6.
- [3] A. Mehta and K. Srinivasan, "RAG-Based LLM Chatbot Using LLaMA-2," *Proceedings of the International Conference on Generative AI Systems*, 2024, pp. 12–19.
- [4] V. Rao, S. Iyer and P. Krishnan, "Localized Open-Source LLM-Aware Retrieval Augmented Generation of Legal Documents: A Case Study on Indian Constitution and Penal Code," *2024 Conference on Legal Informatics and AI*, 2024, pp. 1–7.
- [5] M. Hassan, J. Lee and K. Tan, "Implementing a RAG-Based Chatbot in Industrial Operations: A Real-World Case Study," *IEEE International Conference on Industrial AI Applications*, 2024, pp. 1–9.
- [6] T. Nguyen and P. Roy, "Implementation of RAG Based Question-Answering Application," *International Journal of Artificial Intelligence Research*, vol. 18, no. 3, pp. 45–53, 2024.
- [7] A. Thomas, R. K. Sharma and D. Ghosh, "Generative AI-Powered Conversational Assistant for Document, Web and Repository Interaction and Retrieval," *IEEE Access*, vol. 12, pp. 102345–102360, 2024.
- [8] L. Chen and S. Wong, "Gen AI Driven FAQ Chatbot Using Advanced RAG Architecture for Querying Annual Reports," *2024 International Conference on Financial Technology and AI*, 2024, pp. 1–8.
- [9] Y. Park, M. Ali and R. Gupta, "Enhancing RAG Pipeline Performance with Translation-Based Embedding Strategies for Non-English Documents," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 5, pp. 2201–2213, 2024.
- [10] J. Morales and A. Khan, "Survey of Interactive Document Query Systems Using RAG and LangChain," *ACM Computing Surveys*, vol. 57, no. 2, 2025.
- [11] H. Zhang, P. Liu and D. Roy, "Enhancing Document Retrieval Using AI and Graph-Based RAG Techniques," *2024 International Conference on Knowledge Graphs and Intelligent Systems*, 2024, pp. 1–8.
- [12] S. K. Rao, N. I. Ahmed and R. Kulkarni, "Development of a GI Cancer Diagnostic Chatbot Based on RAG Model," *IEEE International Conference on Biomedical Informatics*, 2024, pp. 1–7.
- [13] A. Menon and K. Das, "Document Query Response Using Microsoft Phi-3-Mini-4K," *2024 International Conference on Generative AI and Applications*, 2024, pp. 1–6.
- [14] R. Sharma, A. Patel and V. Iyer, "DocGenie: A Retrieval-Augmented Generation Chatbot for Interactive PDF Exploration," *Proceedings of the International Conference on Intelligent Information Systems*, 2024, pp. 1–9.

- [15] T. Banerjee and P. Mehra, "Conversational AI Meets Documents: Revolutionizing PDF Interaction with Generative AI," *IEEE International Conference on Human-Centered AI Systems*, 2024, pp. 1–8.
- [16] K. Rahman, J. Choi and S. Li, "Context-Driven Chatbot Development Leveraging Zephyr-7B with RAG for Improved Response Accuracy," *2024 IEEE Conference on Natural Language Processing Systems*, 2024, pp. 1–7.
- [17] M. O. Hassan and L. Fernandez, "Comparative Study of Chatbot Architectures Using LLMs and RAG: Evaluating FAISS-Based Knowledge Retrieval Schemes," *IEEE Access*, vol. 12, pp. 135210–135225, 2024.
- [18] D. Verma, A. Singh and R. K. Jain, "Chatbot for Student Discipline Handbook-Related Queries: A RAG-Based LLM Using Llama-3 Approach," *2025 International Conference on AI in Education*, 2025, pp. 1–6.
- [19] N. Gupta, M. El-Sayed and T. H. Nguyen, "Biomedical Chat Assistant with Personalized Document Reader Using BioMistral and RAG," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 2, pp. 456–468, 2025.
- [20] F. Al-Mansoori and J. Park, "A Comparative Study of Retrieval-Augmented Generation (RAG) Chatbots," *ACM Transactions on Intelligent Systems and Technology*, vol. 16, no. 1, 2025.
- [21] P. Lewis, E. Perez, A. Piktus *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.
- [22] J. Guu, K. Lee, Z. Tung, P. Pasupat and M. Chang, "REALM: Retrieval-Augmented Language Model Pre-Training," *Proceedings of the International Conference on Machine Learning (ICML)*, 2020, pp. 3929–3938.
- [23] K. Izacard and E. Grave, "Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering," *Proceedings of the European Chapter of the Association for Computational Linguistics (EACL)*, 2021, pp. 874–880.
- [24] Y. Liu, M. Ott, N. Goyal *et al.*, "RoBERTa: A Robustly Optimized BERT Pretraining Approach," *arXiv preprint arXiv:1907.11692*, 2019.
- [25] J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [26] J. Johnson, M. Douze and H. Jégou, "Billion-Scale Similarity Search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [27] T. Kwiatkowski, J. Palomaki, O. Redfield *et al.*, "Natural Questions: A Benchmark for Question Answering Research," *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 453–466, 2019.
- [28] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [29] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks," *Proceedings of EMNLP*, 2019, pp. 3982–3992.
- [30] OpenAI, "GPT-4 Technical Report," *arXiv preprint arXiv:2303.08774*, 2023.